Nisandi Disara – 23001951

Assignment: Manual Inspection of a Disk Image

Step 1 - Verify File Integrity (MD5)

Before analysis, the disk image and its source archive were verified against the MD5 hashes published with the assignment to confirm the file was not corrupted or altered.

Command:

md5sum file-system-assignment-image.zip

md5sum file-system-assignment-image.dd

Result:

a984d73c44898cbcfe125ea4034141e9 file-system-assignment-image.zip

86f4c00d26dc664e411bf752434f41d0 file-system-assignment-image.dd

Both hashes matched the values given in the assignment, confirming the image is authentic.

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ md5sum file-system-assignment-image.zip
a984d73c44898cbcfe125ea4034141e9  file-system-assignment-image.zip
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ unzip file-system-assignment-image.zip
Archive:  file-system-assignment-image.zip
  inflating: file-system-assignment-image.dd
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ md5sum file-system-assignment-image.dd
86f4c00d26dc664e411bf752434f41d0  file-system-assignment-image.dd
```

01)

Methodology:

I opened the disk image file-system-assignment-image.dd and inspected the Master Boot Record (MBR) partition table, which is located at offset **0x1BE (446 bytes)** from the start of the disk. The partition table contains four partition entries, each 16 bytes in size.

Command used:

xxd -s 446 -l 64 file-system-assignment-image.dd

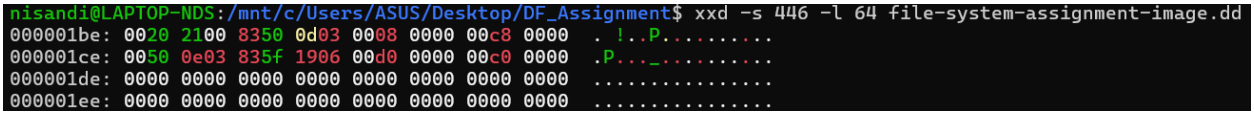
Output:

000001be: 0020 2100 8350 0d03 0008 0000 00c8 0000 . !..P.....

000001ce: 0050 0e03 835f 1906 00d0 0000 00c0 0000 .P..._.....

000001de: 0000 0000 0000 0000 0000 0000 0000 0000 ....

000001ee: 0000 0000 0000 0000 0000 0000 0000 0000 ....



I manually inspected the first two partition entries because the remaining two entries contained only zeros, indicating no partition is defined there.

For each partition entry:

- Byte 4 gives the **partition type code**.
- Bytes 8–11 give the **starting sector (LBA)**, stored in little-endian format.
- Bytes 12–15 give the **partition size in sectors**, stored in little-endian format.

Since the starting sector and size fields are stored in little-endian byte order, I reversed the byte order before converting the hexadecimal values to decimal.

Partition Entry 1:

- Bytes: 00 20 21 00 83 50 0d 03 00 08 00 00 00 c8 00 00
- Partition Type = 83
- Starting Sector Bytes = 00 08 00 00 → little-endian conversion → 00 00 08 00 → 0x800 = **2048**

- Number of Sectors Bytes = 00 c8 00 00 → little-endian conversion → 00 00 c8 00 → 0xC800 = **51200 sectors**

Partition Entry 2:

- Bytes: 00 50 0e 03 83 5f 19 06 00 d0 00 00 00 c0 00 00
- Partition Type = 83
- Starting Sector Bytes = 00 d0 00 00 → little-endian conversion → 00 00 d0 00 → 0xD000 = **53248**
- Number of Sectors Bytes = 00 c0 00 00 → little-endian conversion → 00 00 c0 00 → 0xC000 = **49152 sectors**

Partition	**Type**	**Partition Type Code**	**Starting Sector**	**Size (sectors)**
Partition 1	Linux filesystem	0x83	2048	51200
Partition 2	Linux filesystem	0x83	53248	49152

02)

Methodology:

Using the starting sector and size values obtained in Question 1, I extracted each partition from the disk image using the dd command, then calculated the MD5 hash of each extracted partition file using md5sum.

Partition 1 extraction:

dd if=file-system-assignment-image.dd of=partition1.dd bs=512 skip=2048 count=51200

Output:

51200+0 records in

51200+0 records out

26214400 bytes (26 MB, 25 MiB) copied, 151.713 s, 173 kB/s

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ dd if=file-system-assignment-image.dd of=partition1.dd bs=51
2 skip=2048 count=51200
51200+0 records in
51200+0 records out
26214400 bytes (26 MB, 25 MiB) copied, 151.713 s, 173 kB/s
```

Partition 1 MD5:

md5sum partition1.dd

Output:

1bf402852fda82ac1a46914d339692f4 partition1.dd

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ md5sum partition1.dd
1bf402852fda82ac1a46914d339692f4  partition1.dd
```

Partition 2 extraction:

dd if=file-system-assignment-image.dd of=partition2.dd bs=512 skip=53248 count=49152

Output:

49152+0 records in

49152+0 records out

25165824 bytes (25 MB, 24 MiB) copied, 238.594 s, 105 kB/s

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ dd if=file-system-assignment-image.dd of=partition2.dd bs=51
2 skip=53248 count=49152
49152+0 records in
49152+0 records out
25165824 bytes (25 MB, 24 MiB) copied, 238.594 s, 105 kB/s
```

Partition 2 MD5:

md5sum partition2.dd

Output:

7811442d76f3e4e654907533a6f155a1 partition2.dd

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ md5sum partition2.dd
7811442d76f3e4e654907533a6f155a1  partition2.dd
```

Partition	MD5 Hash
Partition 1	1bf402852fda82ac1a46914d339692f4
Partition 2	7811442d76f3e4e654907533a6f155a1

03)

Methodology:

I opened the extracted partition image partition1.dd and examined the Boot Sector (BPB) to determine the FAT16 layout parameters:

- Bytes per sector = 512 bytes
- Sectors per cluster = 4 sectors
- Reserved sectors = 4 sectors
- Number of FATs = 2
- Number of root directory entries = 512
- Sectors per FAT = 52 sectors

Using these values, I calculated the location of the root directory:

Root Directory Start Sector = Reserved Sectors + (Number of FATs × Sectors per FAT)

= 4 + (2 × 52)

= 108

Root Directory Offset = 108 × 512 = 55296 bytes = 0xD800

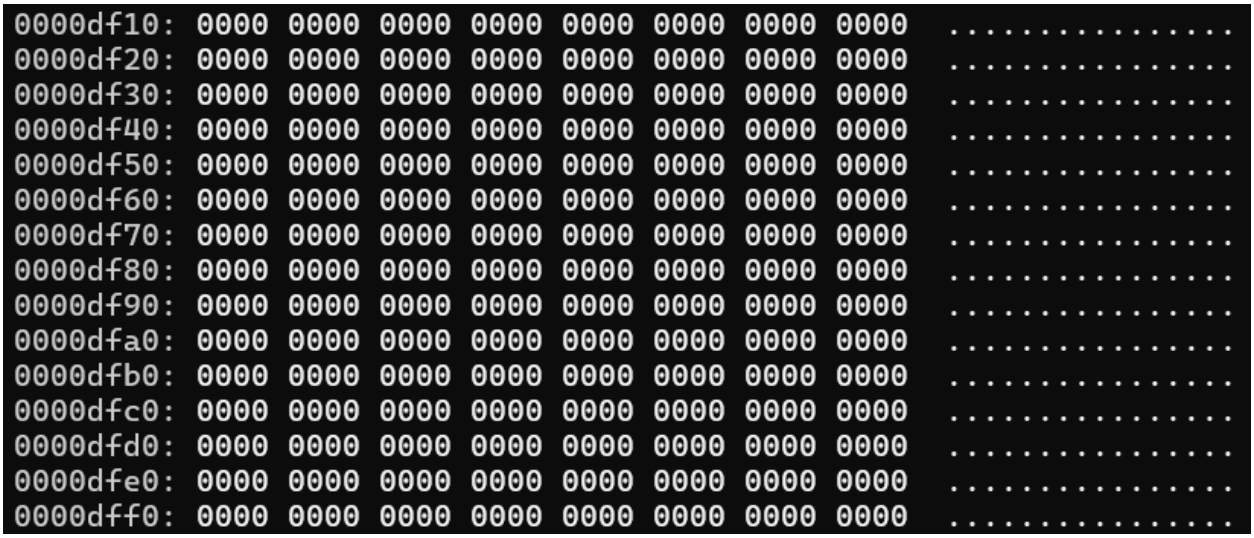
Command used:

xxd -s 55296 -l 2048 partition1.dd



```
0000da60: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000da70: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000da80: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000da90: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000daa0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dab0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dac0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dad0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dae0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000daf0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db00: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db10: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db20: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db30: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db40: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db50: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db60: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db70: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db80: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000db90: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dba0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dbb0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dbc0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dbd0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dbe0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dbf0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc00: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc10: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc20: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc30: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc40: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc50: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc60: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc70: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc80: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dc90: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dca0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
```

```
0000dcb0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dcc0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dcd0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dce0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dcf0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd00: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd10: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd20: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd30: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd40: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd50: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd60: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd70: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd80: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dd90: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dda0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000ddb0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000ddc0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000ddd0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dde0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000ddf0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de00: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de10: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de20: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de30: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de40: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de50: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de60: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de70: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de80: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000de90: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dea0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000deb0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dec0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000ded0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000dee0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000def0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0000df00: 0000 0000 0000 0000 0000 0000 0000 0000  ................
```



Each 32-byte FAT16 directory entry was analyzed. Long File Name (LFN) entries, identified by attribute value 0x0F, were used together with their corresponding short-name entries to identify the actual files. Deleted files were identified by checking the first byte of the short file name entry - a value of 0xE5 indicates the entry has been deleted (only the first character of the name is overwritten, the rest of the name remains readable).

From the root directory, I identified the following entries:

Name	**1st byte**	**Deleted?**
MESSAGE.TXT	4D ('M')	No
~~P~~RINTING	E5	Yes (originally PRINTING)
SPYING.JPG	53 ('S')	No
~~S~~USPECT.JPG	E5	Yes (originally SUSPECT.JPG)
WEAPONS	57 ('W')	No
WHYNOT.JPG	57 ('W')	No

I also identified a subdirectory, WEAPONS, with a starting cluster of 124 (from bytes 7C 00 at the cluster field of its directory entry, little-endian).

To inspect its contents, I calculated the offset of cluster 124:

Data Area Start Sector = Reserved Sectors + FAT1 + FAT2 + Root Directory Sectors

= 4 + 52 + 52 + 32 = 140 sectors

Cluster Sector = Data Area Start + (Cluster Number - 2) × Sectors per Cluster

= 140 + (124 - 2) × 4

= 628 sectors

Byte Offset = 628 × 512 = 321536 bytes (0x4E800)

Command used:

xxd -s 321536 -l 512 partition1.dd

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ xxd -s 321536 -l 512 partition1.dd
0004e800: 2e20 2020 2020 2020 2020 2010 0064 e181  .           ..d..
0004e810: 6751 6751 0000 e181 6751 7c00 0000 0000  gQgQ....gQ|.....
0004e820: 2e2e 2020 2020 2020 2020 2010 0064 e181  ..          ..d..
0004e830: 6751 6751 0000 e181 6751 0000 0000 0000  gQgQ....gQ......
0004e840: 4161 006b 0034 0037 002e 000f 001c 6a00  Aa.k.4.7......j.
0004e850: 7000 6700 0000 ffff ffff 0000 ffff ffff  p.g.............
0004e860: 414b 3437 2020 2020 4a50 4720 0064 e181  AK47    JPG .d..
0004e870: 6751 6751 0000 e181 6751 7d00 5e27 0000  gQgQ....gQ}.^'..
0004e880: 416d 0031 0036 002e 006a 000f 006a 7000  Am.1.6...j...jp.
0004e890: 6700 0000 ffff ffff ffff 0000 ffff ffff  g...............
0004e8a0: 4d31 3620 2020 2020 4a50 4720 0064 e181  M16     JPG .d..
0004e8b0: 6751 6751 0000 e181 6751 8200 3127 0000  gQgQ....gQ..1'..
0004e8c0: 4170 0069 0073 0074 006f 000f 0046 6c00  Ap.i.s.t.o...Fl.
0004e8d0: 2e00 6a00 7000 6700 0000 0000 ffff ffff  ..j.p.g.........
0004e8e0: 5049 5354 4f4c 2020 4a50 4720 0064 e181  PISTOL  JPG .d..
0004e8f0: 6751 6751 0000 e181 6751 8700 4b99 0000  gQgQ....gQ..K...
0004e900: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e910: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e920: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e930: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e940: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e950: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e960: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e970: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e980: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e990: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e9a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e9b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e9c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e9d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e9e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
0004e9f0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
```

Inside the WEAPONS directory, I found the following files, none of which were deleted:

Name	1st byte	Deleted?
AK47.JPG	41 ('A')	No
M16.JPG	4D ('M')	No
PISTOL.JPG	50 ('P')	No

a) File names in Partition 1

Name	Type
message.txt	File
printing	Directory
spying.jpg	File
suspect.jpg	File
weapons	Directory
whynot.jpg	File
ak47.jpg	File
m16.jpg	File
pistol.jpg	File

b) Full file paths from the root directory

File Name	Full Path
message.txt	/message.txt
spying.jpg	/spying.jpg
suspect.jpg	/suspect.jpg
whynot.jpg	/whynot.jpg
ak47.jpg	/weapons/ak47.jpg
m16.jpg	/weapons/m16.jpg
pistol.jpg	/weapons/pistol.jpg

c) Deleted files

Deleted Entry	Type
printing	Directory
suspect.jpg	File

04)

Methodology:

The file message.txt was identified from the FAT16 root directory located at offset 0xD800 in partition1.dd (decoded in Question 3).

From the 32-byte FAT16 directory entry:

- The starting cluster number is stored at bytes 26–27.
- The file size is stored at bytes 28–31.

The extracted values were:

- Starting cluster bytes: 03 00 → little-endian → 0x0003 = **cluster 3**
- File size bytes: 49 00 00 00 → little-endian → 0x00000049 = **73 bytes**

a) File size

Answer: 73 bytes

b) Clusters occupied

The FAT table begins right after the reserved sectors:

FAT1 Offset = Reserved Sectors × Bytes per Sector = 4 × 512 = 2048 bytes (0x800)

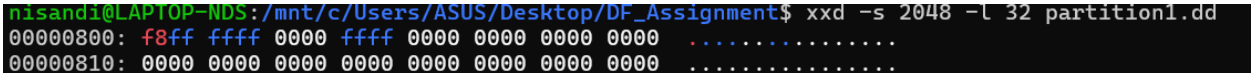
Command used:

xxd -s 2048 -l 32 partition1.dd

Output:

00000800: f8ff ffff 0000 ffff 0000 0000 0000 0000

00000810: 0000 0000 0000 0000 0000 0000 0000 0000



Each FAT16 entry is 2 bytes. Reading the entries in order: entry 0 = F8 FF (media descriptor), entry 1 = FF FF (reserved), entry 2 = 00 00 (free), **entry 3 = FF FF**.

An entry value of 0xFFFF indicates the end of the cluster chain (last cluster of the file, no further clusters allocated).

Answer: message.txt occupies cluster 3 only.

c) Extract file content and MD5

The starting cluster (3) was converted to a byte offset:

Data Area Start Sector = Reserved Sectors + FAT1 + FAT2 + Root Directory

= 4 + 52 + 52 + 32 = 140 sectors

Cluster Sector = Data Area Start + (Cluster Number - 2) × Sectors per Cluster

= 140 + (3 - 2) × 4 = 144 sectors

Byte Offset = 144 × 512 = 73728 bytes (0x12000)

Command used to extract the file:

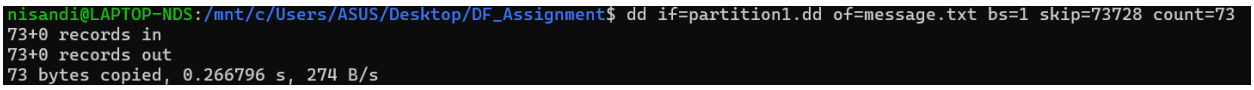
dd if=partition1.dd of=message.txt bs=1 skip=73728 count=73

Output:

73+0 records in

73+0 records out

73 bytes copied, 0.266796 s, 274 B/s



Command used to view content:

cat message.txt

Output:

But, on the instant, came the sweep and flash of Jonathan's great knife.

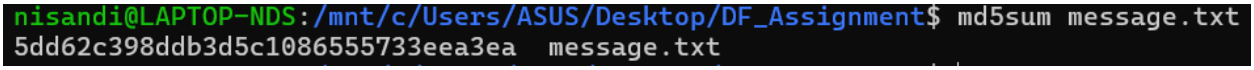


Command used for MD5 hash:

md5sum message.txt

Output:

5dd62c398ddb3d5c1086555733eea3ea message.txt



Item	**Value**
File size	73 bytes
Cluster(s) occupied	Cluster 3
Content	"But, on the instant, came the sweep and flash of Jonathan's great knife."
MD5 hash	5dd62c398ddb3d5c1086555733eea3ea

05)

Methodology:

I examined the NTFS boot sector of partition2.dd to extract the key BPB (BIOS Parameter Block) values.

Command used:

xxd -l 512 partition2.dd

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ xxd -l 512 partition2.dd
00000000: eb52 904e 5446 5320 2020 2000 0208 0000  .R.NTFS    ....
00000010: 0000 0000 00f8 0000 0000 0000 0000 0000  ................
00000020: 0000 0000 8000 8000 ffbf 0000 0000 0000  ................
00000030: 0400 0000 0000 0000 ff0b 0000 0000 0000  ................
00000040: f600 0000 0100 0000 f68c fe1a 124b cb16  .............K..
00000050: 0000 0000 0e1f be71 7cac 22c0 740b 56b4  .......q|.".t.V.
00000060: 0ebb 0700 cd10 5eeb f032 e4cd 16cd 19eb  ......^..2......
00000070: fe54 6869 7320 6973 206e 6f74 2061 2062  .This is not a b
00000080: 6f6f 7461 626c 6520 6469 736b 2e20 506c  ootable disk. Pl
00000090: 6561 7365 2069 6e73 6572 7420 6120 626f  ease insert a bo
000000a0: 6f74 6162 6c65 2066 6c6f 7070 7920 616e  otable floppy an
000000b0: 640d 0a70 7265 7373 2061 6e79 206b 6579  d..press any key
000000c0: 2074 6f20 7472 7920 6167 6169 6e20 2e2e   to try again ..
000000d0: 2e20 0d0a 0000 0000 0000 0000 0000 0000  . ..............
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000100: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000110: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000120: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000130: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000140: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000150: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000160: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000170: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000180: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000190: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001f0: 0000 0000 0000 0000 0000 0000 0000 55aa  ..............U.
```

From the boot sector:

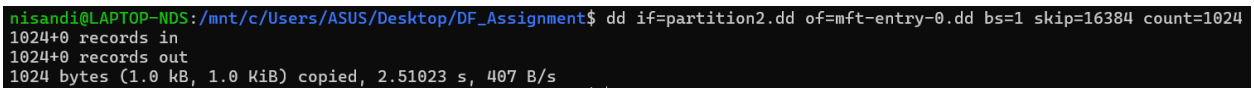
- Filesystem signature (offset 0x03) = "NTFS " — confirms NTFS filesystem
- Bytes per sector = 512 bytes
- Sectors per cluster = 8
- Cluster size = 512 × 8 = **4096 bytes**
- $MFT starting cluster (offset 0x30) = **4**

MFT Offset = Cluster × Cluster Size = 4 × 4096 = 16384 bytes

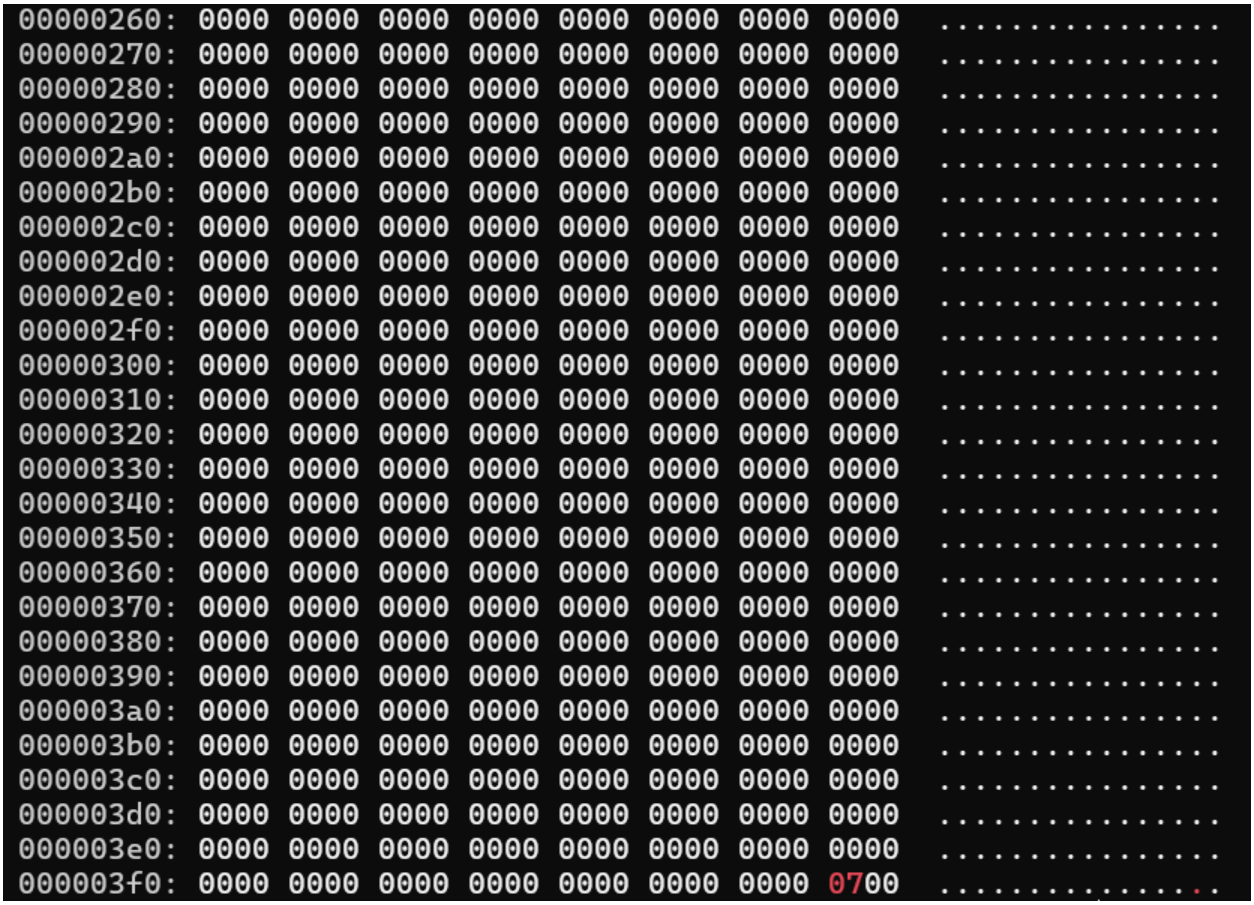
Extracting and verifying MFT Entry 0:

dd if=partition2.dd of=mft-entry-0.dd bs=1 skip=16384 count=1024

xxd mft-entry-0.dd



```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ xxd mft-entry-0.dd
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0100 9801 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0400 0000 0000 0000  ................
00000030: 0700 0000 0000 0000 1000 0000 6000 0000  ............`...
00000040: 0000 1800 0000 0000 4800 0000 1800 0000  ........H.......
00000050: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000060: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000070: 0600 0000 0000 0000 0000 0000 0000 0000  ................
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000090: 0000 0000 0000 0000 3000 0000 6800 0000  ........0...h...
000000a0: 0000 1800 0000 0200 4a00 0000 1800 0100  ........J.......
000000b0: 0500 0000 0000 0500 8034 2a2f 65b4 d601  .........4*/e...
000000c0: 8034 2a2f 65b4 d601 8034 2a2f 65b4 d601  .4*/e....4*/e...
000000d0: 8034 2a2f 65b4 d601 0070 0000 0000 0000  .4*/e....p......
000000e0: 006c 0000 0000 0000 0600 0000 0000 0000  .l..............
000000f0: 0403 2400 4d00 4600 5400 0000 0000 0000  ..$.M.F.T.......
00000100: 8000 0000 4800 0000 0100 4000 0000 0100  ....H.....@.....
00000110: 0000 0000 0000 0000 1200 0000 0000 0000  ................
00000120: 4000 0000 0000 0000 0030 0100 0000 0000  @........0......
00000130: 0014 0100 0000 0000 0014 0100 0000 0000  ................
00000140: 1113 0400 0000 0000 b000 0000 4800 0000  ............H...
00000150: 0100 4000 0000 0300 0000 0000 0000 0000  ..@.............
00000160: 0000 0000 0000 0000 4000 0000 0000 0000  ........@.......
00000170: 0010 0000 0000 0000 1000 0000 0000 0000  ................
00000180: 1000 0000 0000 0000 1101 0200 0000 0000  ................
00000190: ffff ffff 0000 0000 0000 0000 0000 0000  ................
000001a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001f0: 0000 0000 0000 0000 0000 0000 0000 0700  ................
00000200: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000210: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000220: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000230: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000240: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000250: 0000 0000 0000 0000 0000 0000 0000 0000  ................
```



The record began with the ASCII signature FILE, confirming a valid NTFS MFT entry. Each MFT record is exactly 1024 bytes.

Entries 0 - 15 are reserved for NTFS system metadata files ($MFT, $LogFile, $Volume, $AttrDef, $Bitmap, $Boot, $BadClus, $Secure, $Upcase, $Extend, etc.), so they were excluded from user file identification. I began checking from entry 16 onward.

Checking allocation status:

Each MFT record's allocation status is stored at offset **0x16** (2 bytes) of the record header - a value of 0x0001 indicates the record is allocated (in use), while 0x0000 indicates the record is deleted (unallocated).

To efficiently check multiple entries, I used a loop to calculate each entry's byte offset (MFT Offset + Entry Number × 1024) and extract the first 64 bytes of each:

```
for i in 16 17 18 19 20 21 22 23 24 25 26 27 64 65 66 67 68; do
  echo "=== Entry $i ==="
dd if=partition2.dd bs=1 skip=$((16384 + i*1024)) count=64 2>/dev/null | xxd done
```

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ for i in 16 17 18 19 20 21 22 23 24 25 26 27 64 65 66 67 68; do
  echo "=== Entry $i ==="
  dd if=partition2.dd bs=1 skip=$((16384 + i*1024)) count=64 2>/dev/null | xxd
done
=== Entry 16 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 1000 0000 3800 0000 8800 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0100 0000 0000 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 17 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 1100 0000 3800 0000 8800 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0100 0000 0000 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 18 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 1200 0000 3800 0000 8800 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0100 0000 0000 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 19 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 1300 0000 3800 0000 8800 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0100 0000 0000 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 20 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 1400 0000 3800 0000 8800 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0100 0000 0000 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 21 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 1500 0000 3800 0000 8800 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0100 0000 0000 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 22 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 1600 0000 3800 0000 8800 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0100 0000 0000 0000  ................
```

```
=== Entry 23 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 1700 0000 3800 0000 8800 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0100 0000 0000 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 24 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0d00 7002 0000 0004 0000  ....8...p.......
00000020: 0000 0000 0000 0000 0400 0000 1800 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 6000 0000  ............`...
=== Entry 25 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0d00 5801 0000 0004 0000  ....8...X.......
00000020: 0000 0000 0000 0000 0300 0000 1900 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 6000 0000  ............`...
=== Entry 26 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0d00 6001 0000 0004 0000  ....8...`.......
00000020: 0000 0000 0000 0000 0300 0000 1a00 0000  ................
00000030: 0200 0000 0000 0000 1000 0000 6000 0000  ............`...
=== Entry 27 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0000 3800 0000 4000 0000 0004 0000  ....8...@.......
00000020: 0000 0000 0000 0000 0000 0000 1b00 0000  ................
00000030: 0200 0000 0000 0000 ffff ffff 0000 0000  ................
=== Entry 64 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0100 c001 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0400 0000 4000 0000  ............@...
00000030: 5900 0000 0000 0000 1000 0000 4800 0000  Y...........H...
=== Entry 65 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0100 a801 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0400 0000 4100 0000  ............A...
00000030: b301 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 66 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0100 a801 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0400 0000 4200 0000  ............B...
00000030: 4b07 0000 0000 0000 1000 0000 4800 0000  K...........H...
```

```
=== Entry 67 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0100 c001 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0400 0000 4300 0000  ............C...
00000030: 8b01 0000 0000 0000 1000 0000 4800 0000  ............H...
=== Entry 68 ===
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0100 b801 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0400 0000 4400 0000  ............D...
00000030: 2d01 0000 0000 0000 1000 0000 4800 0000  -...........H...
```

Findings:

Entry Range	Allocation Flag	Status	Notes
16–23	00 00	Deleted	Unallocated records
24	0d 00	Allocated	System file — $Quota
25	0d 00	Allocated	System file — $ObjId
26	0d 00	Allocated	System file — $Reparse
27–63	00 00	Deleted	Unallocated records
64–68	01 00	**Allocated**	**User files**

Extracting filenames from the allocated user file entries:

For each allocated entry (64–68), the $FILE_NAME attribute (identifier 30 00 00 00) begins at offset 128 within the record (since the 1st attribute, $STANDARD_INFORMATION, starts at offset 56 and is 72 bytes long: 56 + 72 = 128).

```
for i in 64 65 66 67 68; do
  echo "=== Entry $i filename ==="
  dd if=partition2.dd bs=1 skip=$((16384 + i*1024 + 128)) count=136 2>/dev/null | xxd
done
```

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ for i in 64 65 66 67 68; do
  echo "=== Entry $i filename ==="
  dd if=partition2.dd bs=1 skip=$((16384 + i*1024 + 128)) count=136 2>/dev/null | xxd
done
=== Entry 64 filename ===
00000000: 3000 0000 8800 0000 0000 0000 0000 0300  0...............
00000010: 7000 0000 1800 0100 0500 0000 0000 0500  p...............
00000020: c779 4d26 21b5 d601 c779 4d26 21b5 d601  .yM&!....yM&!...
00000030: c779 4d26 21b5 d601 c779 4d26 21b5 d601  .yM&!....yM&!...
00000040: 00b0 0200 0000 0000 0000 0000 0000 0000  ................
00000050: 2000 0000 0000 0000 1700 4100 6c00 6900   .........A.l.i.
00000060: 6300 6500 2d00 6900 6e00 2d00 5700 6f00  c.e.-.i.n.-.W.o.
00000070: 6e00 6400 6500 7200 6c00 6100 6e00 6400  n.d.e.r.l.a.n.d.
00000080: 2e00 7400 7800 7400                      ..t.x.t.
=== Entry 65 filename ===
00000000: 3000 0000 7000 0000 0000 0000 0000 0300  0...p...........
00000010: 5800 0000 1800 0100 0500 0000 0000 0500  X...............
00000020: 21e2 4d26 21b5 d601 21e2 4d26 21b5 d601  !.M&!...!.M&!...
00000030: 21e2 4d26 21b5 d601 21e2 4d26 21b5 d601  !.M&!...!.M&!...
00000040: 0080 0d00 0000 0000 0000 0000 0000 0000  ................
00000050: 2000 0000 0000 0000 0b00 4400 7200 6100   .........D.r.a.
00000060: 6300 7500 6c00 6100 2e00 7400 7800 7400  c.u.l.a...t.x.t.
00000070: 5000 0000 6800 0000 0000 0000 0000 0100  P...h...........
00000080: 5000 0000 1800 0000                      P.......
=== Entry 66 filename ===
00000000: 3000 0000 7000 0000 0000 0000 0000 0300  0...p...........
00000010: 5400 0000 1800 0100 0500 0000 0000 0500  T...............
00000020: 9f2c 5026 21b5 d601 9f2c 5026 21b5 d601  .,P&!....,P&!...
00000030: 9f2c 5026 21b5 d601 9f2c 5026 21b5 d601  .,P&!....,P&!...
00000040: 0040 3a00 0000 0000 0000 0000 0000 0000  .@:.............
00000050: 2000 0000 0000 0000 0900 7000 6800 6f00   .........p.h.o.
00000060: 7400 6f00 2e00 6a00 7000 6700 1800 0000  t.o...j.p.g.....
00000070: 5000 0000 6800 0000 0000 0000 0000 0100  P...h...........
00000080: 5000 0000 1800 0000                      P.......
=== Entry 67 filename ===
00000000: 3000 0000 8800 0000 0000 0000 0000 0300  0...............
00000010: 7000 0000 1800 0100 0500 0000 0000 0500  p...............
00000020: 6089 5626 21b5 d601 6089 5626 21b5 d601  `.V&!...`.V&!...
00000030: 6089 5626 21b5 d601 6089 5626 21b5 d601  `.V&!...`.V&!...
00000040: 0040 0c00 0000 0000 0000 0000 0000 0000  .@..............
```

```
=== Entry 68 filename ===
00000000: 3000 0000 8000 0000 0000 0000 0000 0300  0...............
00000010: 6800 0000 1800 0100 0500 0000 0000 0500  h...............
00000020: 60be 5726 21b5 d601 60be 5726 21b5 d601  `.W&!...`.W&!...
00000030: 60be 5726 21b5 d601 60be 5726 21b5 d601  `.W&!...`.W&!...
00000040: 0050 0900 0000 0000 0000 0000 0000 0000  .P..............
00000050: 2000 0000 0000 0000 1300 5300 6800 6500   .........S.h.e.
00000060: 7200 6c00 6f00 6300 6b00 2d00 4800 6f00  r.l.o.c.k.-.H.o.
00000070: 6c00 6d00 6500 7300 2e00 7400 7800 7400  l.m.e.s...t.x.t.
00000080: 5000 0000 6800 0000                      P...h...
```

The Unicode (UTF-16LE) filenames decoded from the $FILE_NAME attribute of each entry were:

MFT Entry	Filename
64	Alice-in-Wonderland.txt
65	Dracula.txt
66	photo.jpg
67	Pride-and-Prejudice.txt
68	Sherlock-Holmes.txt

a) File names in Partition 2

- Alice-in-Wonderland.txt
- Dracula.txt
- photo.jpg
- Pride-and-Prejudice.txt
- Sherlock-Holmes.txt

b) Full file paths from the root directory

All five files are standalone MFT entries with no parent subdirectory, meaning they reside directly in the root directory of Partition 2.

File Name	Full Path
Alice-in-Wonderland.txt	/Alice-in-Wonderland.txt
Dracula.txt	/Dracula.txt
photo.jpg	/photo.jpg
Pride-and-Prejudice.txt	/Pride-and-Prejudice.txt
Sherlock-Holmes.txt	/Sherlock-Holmes.txt

c) Deleted files

A full attribute walkthrough of these deleted entries (16-23, 27-63) was not performed, as they were confirmed unallocated (allocation flag 00 00) and extracting their file names was not required to answer the question of which *user* files are currently deleted.

06)

Methodology:

The file photo.jpg is stored in **MFT Entry 66** of Partition 2 (identified in Question 5). I extracted the full 1024-byte MFT record and analyzed its attributes.

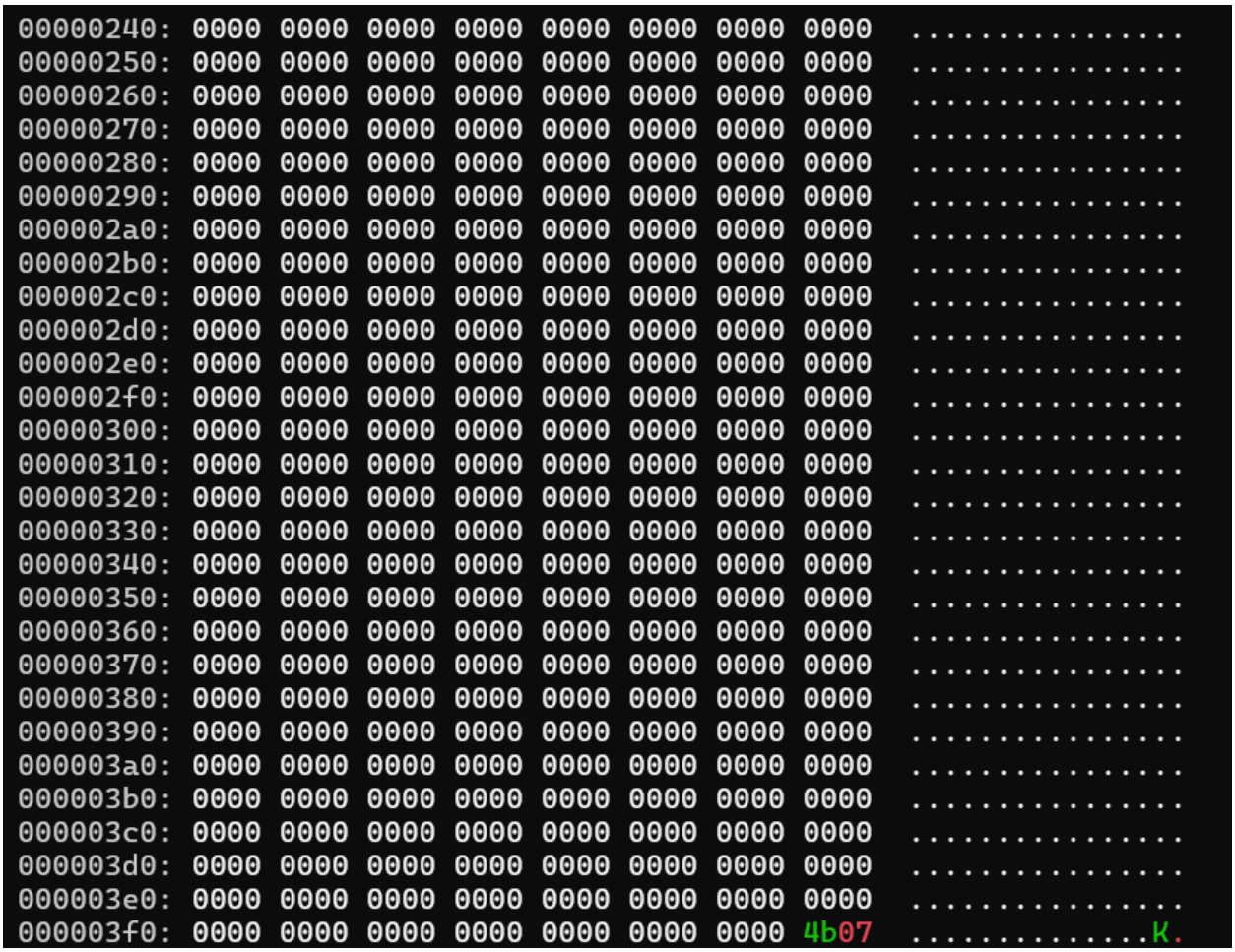
Command used:

dd if=partition2.dd of=mft-entry-66.dd bs=1 skip=$((16384 + 66*1024)) count=1024

xxd mft-entry-66.dd

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ dd if=partition2.dd of=mft-entry-66.dd bs=1 skip=$((16384 + 66*1024)) count=1024
1024+0 records in
1024+0 records out
1024 bytes (1.0 kB, 1.0 KiB) copied, 2.33881 s, 437 B/s
```

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ xxd mft-entry-66.dd
00000000: 4649 4c45 3000 0300 0000 0000 0000 0000  FILE0...........
00000010: 0100 0100 3800 0100 a801 0000 0004 0000  ....8...........
00000020: 0000 0000 0000 0000 0400 0000 4200 0000  ............B...
00000030: 4b07 0000 0000 0000 1000 0000 4800 0000  K...........H...
00000040: 0000 0000 0000 0000 3000 0000 1800 0000  ........0.......
00000050: 9f2c 5026 21b5 d601 1f83 5626 21b5 d601  .,P&!.....V&!...
00000060: 1f83 5626 21b5 d601 9f2c 5026 21b5 d601  ..V&!....,P&!...
00000070: 2000 0000 0000 0000 0000 0000 0000 0000   ...............
00000080: 3000 0000 7000 0000 0000 0000 0000 0300  0...p...........
00000090: 5400 0000 1800 0100 0500 0000 0000 0500  T...............
000000a0: 9f2c 5026 21b5 d601 9f2c 5026 21b5 d601  .,P&!....,P&!...
000000b0: 9f2c 5026 21b5 d601 9f2c 5026 21b5 d601  .,P&!....,P&!...
000000c0: 0040 3a00 0000 0000 0000 0000 0000 0000  .@:.............
000000d0: 2000 0000 0000 0000 0900 7000 6800 6f00   .........p.h.o.
000000e0: 7400 6f00 2e00 6a00 7000 6700 1800 0000  t.o...j.p.g.....
000000f0: 5000 0000 6800 0000 0000 0000 0000 0100  P...h...........
00000100: 5000 0000 1800 0000 0100 0480 1400 0000  P...............
00000110: 2400 0000 0000 0000 3400 0000 0102 0000  $.......4.......
00000120: 0000 0005 2000 0000 2002 0000 0102 0000  .... ... .......
00000130: 0000 0005 2000 0000 2002 0000 0200 1c00  .... ... .......
00000140: 0100 0000 0003 1400 ff01 1f00 0101 0000  ................
00000150: 0000 0001 0000 0000 8000 0000 4800 0000  ............H...
00000160: 0100 4000 0000 0200 0000 0000 0000 0000  ..@.............
00000170: a303 0000 0000 0000 4000 0000 0000 0000  ........@.......
00000180: 0040 3a00 0000 0000 b13a 3a00 0000 0000  .@:......::.....
00000190: b13a 3a00 0000 0000 22a4 0303 0f00 0000  .::.....".......
000001a0: ffff ffff 0000 0000 ffff ffff 0000 0000  ................
000001b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001f0: 0000 0000 0000 0000 0000 0000 0000 4b07  ..............K.
00000200: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000210: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000220: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000230: 0000 0000 0000 0000 0000 0000 0000 0000  ................
```



Attribute walkthrough:

- **1st Attribute – $STANDARD_INFORMATION** (offset 56): size = 48 00 00 00 = 72 bytes → next attribute at 56+72 = 128
- **2nd Attribute – $FILE_NAME** (offset 128): size = 70 00 00 00 = 112 bytes → filename confirmed as **photo.jpg** → next attribute at 128+112 = 240
- **3rd Attribute – $SECURITY_DESCRIPTOR** (offset 240): size = 68 00 00 00 = 104 bytes → next attribute at 240+104 = 344
- **4th Attribute – $DATA** (offset 344): identifier 80 00 00 00

Command used to inspect the $DATA attribute:

xxd -s 344 -l 72 mft-entry-66.dd

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ xxd -s 344 -l 72 mft-entry-66.dd
00000158: 8000 0000 4800 0000 0100 4000 0000 0200  ....H.....@.....
00000168: 0000 0000 0000 0000 a303 0000 0000 0000  ................
00000178: 4000 0000 0000 0000 0040 3a00 0000 0000  @........@:.....
00000188: b13a 3a00 0000 0000 b13a 3a00 0000 0000  .:.......:.....
00000198: 22a4 0303 0f00 0000                      ".......
```

At offset +0x08 of the $DATA attribute, the value is 01, indicating the attribute is **non-resident** — the actual file content is stored in clusters outside the MFT record.

a) File size

The real file size is stored at offset +0x30 of the $DATA attribute:

Value: b1 3a 3a 00 00 00 00 00 → little-endian → 0x003A3AB1

Answer: File size = 3,816,113 bytes

b) Cluster(s) occupied

The allocated size is stored at offset +0x28:

Value: 00 40 3a 00 00 00 00 00 → little-endian → 0x003A4000 = 3,817,472 bytes

Number of clusters = 3,817,472 / 4,096 = 932 clusters

The data runlist begins at offset +0x40:

Runlist bytes: 22 A4 03 03 0F 00

Header 22 indicates: length field = 2 bytes, offset field = 2 bytes.

- Length field: A4 03 → little-endian → 0x03A4 = **932 clusters**
- Starting cluster offset: 03 0F → little-endian → 0x0F03 = **cluster 3843**

Ending cluster = 3843 + 932 - 1 = 4774

Answer: photo.jpg occupies clusters 3843 to 4774 (932 clusters, starting cluster 3843)

c) Extract content + MD5

Byte offset = Starting Cluster × Cluster Size = 3843 × 4096 = 15,740,928 bytes

Command used to extract the exact file content:

dd if=partition2.dd of=photo.jpg bs=1 skip=15740928 count=3816113

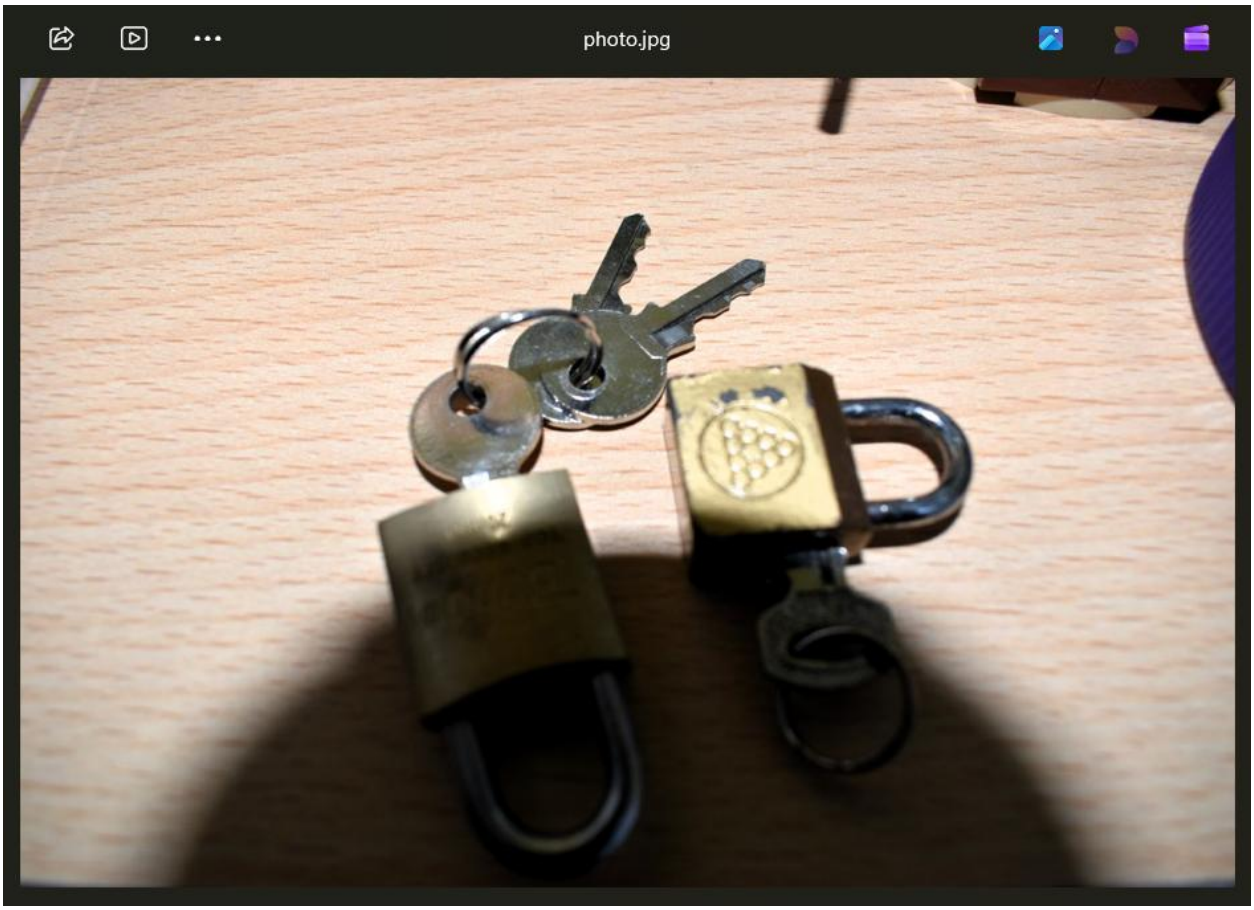
```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ dd if=partition2.dd of=photo.jpg bs=1 skip=15740928 count=3816113
3816113+0 records in
3816113+0 records out
3816113 bytes (3.8 MB, 3.6 MiB) copied, 7992.05 s, 477 B/s
```

Command used to calculate MD5:

md5sum photo.jpg

```
nisandi@LAPTOP-NDS:/mnt/c/Users/ASUS/Desktop/DF_Assignment$ md5sum photo.jpg
492caf78be724608cacf4af401feaeb6  photo.jpg
```

Item	**Value**
File size	3,816,113 bytes
Clusters occupied	3843 – 4774 (932 clusters)
MD5 hash	492caf78be724608cacf4af401feaeb6


photo.jpg